Retrieval-Augmented Generation in Specialized Domains: A Survey

1. Abstract & Introduction

Abstract: Large Language Models (LLMs) often suffer from factual hallucinations and knowledge cutoff limitations. Retrieval-Augmented Generation (RAG) addresses these vulnerabilities by fetching verified evidence from a dynamic vector database. In this empirical study, we evaluate hybrid retrieval mechanisms combining dense sentence embeddings with lexical BM25 matching across technical corpora.

Our experimental pipeline utilizes all-MiniLM-L6-v2 embeddings generating 384-dimensional dense vectors stored within ChromaDB. Baseline evaluations show that dense retrieval alone achieves 78.4% Recall@5, whereas hybrid retrieval enhances Recall@5 to 89.2% on domain queries.

2. Methodology and Chunking Analysis

Document chunking strategy plays a decisive role in retrieval precision. When text chunks are oversized (> 1000 characters), embedding dilution occurs, lowering similarity scores for fine-grained factual queries.

Conversely, undersized chunks (< 150 characters) fail to encapsulate full semantic context. Our optimal configuration employs 500-character chunks with an 80-character sliding overlap, preserving sentence boundaries and mitigating contextual boundary loss.

3. Experimental Results and Conclusion

The primary research finding is that prompt injection defense can be strengthened by explicitly demarcating external passages using XML tags and enforcing strict instruction grounding. Models instructed with source verification exhibited a 92% reduction in ungrounded assertions.

Conclusion: Source-grounded RAG architectures offer enterprise-grade auditability while drastically minimizing generative hallucinations across specialized knowledge bases.